WORKED SOLUTION

STEP I 1988, Question 10

Mechanics • Projectile Motion

STEP I 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q10

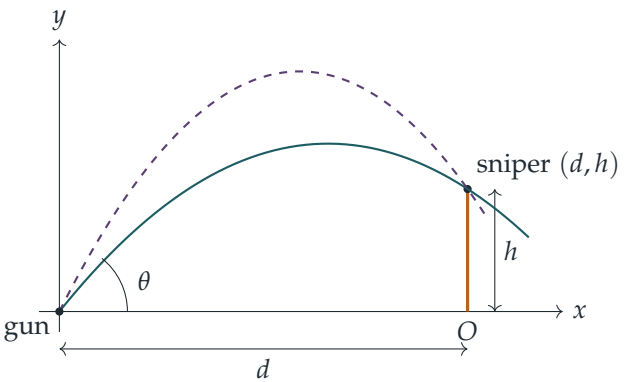
The key idea

Turn the question round. Put the gun at horizontal distance d from the base of the tree and ask: is there *any* angle of elevation for which a bullet with speed v passes through the top of the tree? The trajectory equation, written in terms of $\tan\theta$, is a quadratic in $\tan\theta$, so the answer is "yes" exactly when that quadratic has a real root. The discriminant condition then turns out to be an inequality on d alone.

Solution

The model

We treat the bullet as a particle moving under gravity alone (no air resistance), fired from ground level with speed v at an angle θ above the horizontal, where $0 < \theta \leq \frac{\pi}{2}$. Since the gun's position is unknown, let it be at distance $d \geq 0$ from the base O of the tree, and take axes with origin at the gun, x horizontal towards the tree and y vertically up. The top of the tree is then the point (d, h).



Hitting the sniper

After time t the bullet is at $x = vt\cos\theta$, $y = vt\sin\theta - \frac{1}{2}gt^2$. If $\theta < \frac{\pi}{2}$ we eliminate $t = x/(v\cos\theta)$ and use $\sec^2\theta = 1 + \tan^2\theta$ to get the trajectory

$$y = x\tan\theta - \frac{gx^2}{2v^2}(1 + \tan^2\theta).$$

The bullet hits the sniper if and only if (d, h) lies on this curve. Suppose first that $d > 0$. Writing $T = \tan\theta$ and $k = \dfrac{gd^2}{2v^2} > 0$, the condition is

$$kT^2 - dT + (h + k) = 0.$$

Every real number T is $\tan\theta$ for exactly one $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Can a root be negative, though? The product of the roots is $(h + k)/k > 0$ and their sum is $d/k > 0$, so any real roots are both positive and give genuine elevations $0 < \theta < \frac{\pi}{2}$. Hence some elevation works exactly when this quadratic has a real root, that is, when its discriminant is non-negative:

$$\begin{aligned} d^2 - 4k(h+k) \geq 0 &\iff d^2 \geq \frac{2gd^2}{v^2}\left(h + \frac{gd^2}{2v^2}\right) \\ &\iff v^4 \geq 2ghv^2 + g^2d^2 \qquad \text{(dividing by } d^2 > 0\text{, multiplying by } v^4\text{)} \\ &\iff d^2 \leq \frac{v^2}{g^2}(v^2 - 2gh). \end{aligned}$$

Ask yourself What if the gun was right at the base of the tree, so that $d = 0$ and the trajectory equation above is no use?

Then the bullet must have been fired vertically ($\theta = \frac{\pi}{2}$), and it reaches height h if and only if $v^2 \geq 2gh$, by $v^2 = u^2 - 2gs$ with final speed at least zero. This is the same condition with $d = 0$, so the inequality covers every case.

Since $d \geq 0$, taking the non-negative square root gives

$$d \leq \frac{v}{g}\sqrt{v^2 - 2gh}.$$

So the gun was at a point on the horizontal ground whose distance from the base of the tree is at most $(v/g)\sqrt{v^2 - 2gh}$: it lay within (or, in the extreme case, on) the circle centred on the base of the tree with that radius, as required. The square root is real because we know the shot happened: with $d = 0$ the inequality forces $v^2 \geq 2gh$, which simply says the bullet has enough energy to climb to height h.

The condition is also sufficient. From any point inside the circle the quadratic has two real positive roots, so there are two possible elevations (a flatter and a steeper shot, as in the figure), and on the circle itself they merge into one.

Checks

- With $h = 0$ the radius is v^2/g, the familiar maximum range on level ground. As $v^2 \to 2gh$ the radius shrinks to 0: the gun can only just reach the sniper by firing straight up from the base.
- Units: $(v/g)\sqrt{v^2}$ has units $(\mathrm{m\,s^{-1}})^2/(\mathrm{m\,s^{-2}}) = \mathrm{m}$.
- Numerically, with $g = 9.8$, $v = 20$ and $h = 8$ the radius is about 31.8 m. Stepping the motion forward from $d = 31.8\,\mathrm{m}$ at the single permitted elevation, $\tan\theta = v^2/(gd)$, brings the bullet to height 8.000 m at the tree. A scan over all elevations confirms that points a few per cent outside the circle can't be reached and points inside can.

Common slips

- Using the maximum range v^2/g or the maximum height alone. The sniper is at height h, so both coordinates matter at once.
- Forgetting to check that the roots for $\tan\theta$ give elevations above the horizontal, or ignoring the vertical shot $d = 0$.

Beyond STEP

For fixed v, the points reachable from the gun fill the region below the parabola $y = \dfrac{v^2}{2g} - \dfrac{gx^2}{2v^2}$, the *envelope* of the family of trajectories (the "parabola of safety"). Our discriminant condition is exactly $h \leq$ this envelope at $x = d$. Envelopes of families of curves appear again in the calculus of variations and the theory of first-order differential equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with this kind of care.

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